

ECON 100 - Introduction to Economics

Homework 4

- Define any variables you need to answer the problems.
- You may use any material from this course (including other students). You may use Gen AI to help check your work or understanding, but you should not copy-paste solutions or answers obtained from Chat GPT/AI.
- Define any variables you need to answer the problems.
- Keep your short-answer explanations short but clear. Your goal is to convince a skeptical grader that you understand the relevant concepts well enough to answer the question you are given.
- The questions on this homework sum to 60.
- Please be sure to write down the name(s) of anyone else you worked with on this homework (either at the top or under your name). Place a star next to your name OR underline OR some way to make it easy for me to tell which homework is whose.
- Bring any and all questions to office/student hours!

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1. (2 points) What is your name?
 2. (2 points) What section are you in? (Either section number or what time your class meets)

1 True/False Questions

Indicate “T” rue or “F” alse for each of the following statements or claims. If you correctly explain why a statement or claim is false, I will give you a bonus point up to a 100% on this homework.

3. (2 points) Suppose the market for pharmaceuticals is in equilibrium. **Claim:** If drug makers exit the market AND consumers’ demand for pharmaceuticals increases at every price point then equilibrium quantity will go up but the effect on equilibrium price is unclear.
4. (2 points) A shift in supply will lead to a greater change in equilibrium quantity if the demand curve is more inelastic.
5. (2 points) If the price in a market is below the equilibrium price, the price will increase as a result of the invisible hand until the price reaches the equilibrium price.
6. (2 points) Our supply and demand model predicts the price and quantity in any given market, and it is not guaranteed that the price and quantity will actually occur in that market.
7. (2 points) A movement along the demand curve could result from consumers experiencing a change in their income.

2 Multiple Choice Questions

Choose the best answer for each question. There is ONLY one correct answer for each question. No explanation necessary.

8. (2 points) Suppose that market demand for eggs in a neighborhood is given by the equation $Q^D = 24 - 12p$. The neighborhood has 4 households which produce eggs. Each egg producing household has a supply curve $Q^S = 3p$. Assume that the households cannot set the price of eggs and so take the market price as given. What is the equilibrium price and quantity of eggs in the neighborhood?
- A. $Q=4.8, P=1.60$.
 - B. $Q=12, P=1.6$.
 - C. $Q=4.8, P=1$.
 - D. $Q=12, P=1$.
9. (2 points) Suppose more neighbors decide to start selling eggs, so the supply of eggs triples. What is the new equilibrium price?
- A. $P=0.5$.
 - B. $P=0.75$
 - C. $P=1.5$
 - D. $P=2$.
10. (2 points) Suppose we return to our original problem. $Q^D = 24 - 12P$, and $Q^S = 12P$. If the neighborhood HOA set a price ceiling of 2, which of the following correctly describes how the price ceiling would affect the market?
- A. There would be a shortage.
 - B. There would be a surplus.
 - C. The market would remain unchanged.
 - D. The price would increase but the quantity sold would remain unchanged.
11. (2 points) What if instead the residents convince the HOA to set a price floor of \$2?
- A. There would be a shortage.
 - B. There would be a surplus.
 - C. The market would remain unchanged.
 - D. The price would increase but the quantity sold would remain unchanged.
12. (2 points) Demand for gourmet coffee is given by $Q^D = 100 - P$ and supply is given by $Q^S = 4P$. Now suppose the number of coffee suppliers quadruples. How would the equilibrium price and quantity change?
- A. Equilibrium price decreases, quantity decreases.
 - B. Equilibrium price decreases, quantity increases.
 - C. Equilibrium price increases, quantity decreases.
 - D. Equilibrium price increases, quantity increases.

3 Short Answer Questions

These questions all require math, diagrams, and/or written explanations. Remember you are trying to convince me you understand the why and how of what you are doing, not simply that you are able to get the numerical/graphical answer correct. Include specific concepts from class in your written explanations for full credit.

13. **Tariffs & International Trade.** Suppose you are asked to predict the impacts of a new tariff on steel on domestic consumers and producers for the US. A tariff is a tax added to each ton of imported steel. You have figured out that domestic demand for steel is given by $Q^D(P) = 1500 - 10P$. The domestic supply of steel is given by $Q^S = 50P - 1500$.
- (a) (6 points) Draw a supply and demand diagram for the domestic steel market. Be sure to label your axes and curves. Also indicate the equilibrium supply and demand. No explanation necessary.
 - (b) (4 points) Calculate the equilibrium price and quantity in the market. No explanation necessary.
 - (c) (4 points) Suppose the world price of steel is the same as the equilibrium price you just found above. How much steel would the US import at this price?
 - (d) (6 points) Suppose instead the world price of steel was \$40. On your supply-demand diagram, indicate this world price. How much steel would the US import now? Show US imports graphically and all calculations to arrive at your answer.
 - (e) (6 points) What would happen if the world price of steel was \$60? Use both your supply-demand diagram and math/calculations.
 - (f) (10 points) Based on this problem/what we have discussed in class, what are some pros and cons of tariffs? What are your thoughts on whether or not tariffs should be in place?

4 Ungraded Questions

These questions will not be graded, and solutions may be posted for select questions but there is no guarantee. See office hours for specific questions!

14. **How Does a New College Affect Housing Prices?** Suppose there exists a small town in Upstate New York that does not have a college. Let's call it New Hartford. Suppose the supply of one-bedroom apartment units is given by $Q^S = 100P$, where P is the monthly rent in thousands of dollars. The demand for apartment units is given by $Q^D = 1000 - 100P$.
- (a) (0 points) What is the current equilibrium price and quantity of apartments rented in New Hartford? Show all calculations.
 - (b) (0 points) Draw a supply-demand diagram and label supply, demand, and equilibrium price and quantity. Be sure to label your axes.
 - (c) (0 points) Suppose New Hartford receives a grant from Albany to build 1000 units at no cost to existing landlords and requires no additional cost for the landlords to rent out these apartments. Albany pays all costs (e.g., electricity, heat, water, etc).
 - (d) (0 points) On a new supply-demand diagram, plot the new quantity supplied and original quantity demanded equation. What do you notice?
 - (e) (0 points) What does this imply about the new equilibrium price and quantity after these 1000 units are built?
 - (f) (0 points) Does this implication align with our usual finding of "supply increases, so quantity increases and price decreases"? Why or why not?

15. **How Might the Price of AI Change in the Future?** Suppose that on average, an individual's demand for AI tokens per month is given by $Q^D = 2,000,000 - 100,00P$, where P is given in dollars per month. Suppose that an AI company's quantity of tokens supplied to each individual per month is capped at $Q^S = 1,000,000$.
- (a) (0 points) Calculate the equilibrium price and quantity per person per month in terms of number of tokens. Show all calculations.
 - (b) (0 points) Draw a supply-demand diagram for this market. Be sure to label your axes, curves, and the equilibrium price and quantity.
 - (c) (0 points) Before doing any math, how would you expect the equilibrium price and quantity to change if the AI company increased their number of servers such that they could supply 1,500,000 tokens to each individual per month?
 - (d) (0 points) Shifting the appropriate curve(s), show the new equilibrium price and quantity after the increase in number of servers.
 - (e) (0 points) Calculate the new equilibrium price and quantity.
 - (f) (0 points) Did the math match your predictions? Explain.

16. **How Does Elasticity Affect Changes in Equilibrium Price and Quantity?** To see the effect of elasticity on equilibrium price and quantity, we are going to make $3 \times 8 = 24$ graphs. Each graph will be a supply-demand diagram with an initial equilibrium and an equilibrium price and quantity after we shift one or both curves.
- (a) (0 points) **Equally elastic** Draw 8 supply and demand diagram where supply and demand are approximately equally elastic. On each graph show the effect of the following on equilibrium price and quantity:
1. Demand shifts outward.
 2. Demand shifts inward.
 3. Supply shifts outward.
 4. Supply shifts inward.
 5. Demand and supply shift outward.
 6. Demand and supply shift inward.
 7. Demand shifts outward, supply shifts inward.
 8. Demand shifts inward, supply shifts outward.
- (b) (0 points) **Demand relatively elastic** Draw 8 supply and demand diagram where demand is more elastic than supply. On each graph show the effect of the following on equilibrium price and quantity:
1. Demand shifts outward.
 2. Demand shifts inward.
 3. Supply shifts outward.
 4. Supply shifts inward.
 5. Demand and supply shift outward.
 6. Demand and supply shift inward.
 7. Demand shifts outward, supply shifts inward.
 8. Demand shifts inward, supply shifts outward.
- (c) (0 points) **Demand relatively inelastic** Draw 8 supply and demand diagram where demand is more inelastic than supply. On each graph show the effect of the following on equilibrium price and quantity:
1. Demand shifts outward.
 2. Demand shifts inward.
 3. Supply shifts outward.
 4. Supply shifts inward.
 5. Demand and supply shift outward.
 6. Demand and supply shift inward.
 7. Demand shifts outward, supply shifts inward.
 8. Demand shifts inward, supply shifts outward.
- (d) (0 points) What are 5 takeaways from these graphs?